Machine Learning and Pattern Recognition

Probability Distributions

Dr. Kewei Li

Department of Digital Engineering
Long Island University

February 21, 2025

Introduction

- Introduction
- Binary Variables
- Multinomial Variables
- Gaussian Distribution

Outline

- Introduction
- Binary Variables
- Multinomial Variables
- Gaussian Distribution

Introduction

- We will study some examples of probability distributions and their properties
- One application of the distributions is to model a random variable $\mathbf{x}$ with a distribution $p(\mathbf{x})$, given some training data (density estimation)
- The density estimation problem is ill-posed, model selection is a central issue in pattern recognition

Outline

- Introduction
- Binary Variables
- Multinomial Variables
- Gaussian Distribution

Binary Variables

- Consider a single binary random variable $x \in \{0, 1\}$
- It might describe the outcome of flipping a coin, with $x = 1$ representing 'heads', and $x = 0$ representing the 'tails'
- For a damaged coin, $p(x = 1|\mu) \neq p(x = 0|\mu)$.
- We denote the probability of $x = 1$ by the parameter $\mu \in [0, 1]$, thus

$$p(x = 1|\mu) = \mu$$

- From which it follows that $p(x = 0|\mu) = 1 - \mu$.

Binary Variables

- Thus, we write the probability distribution of x as

$$\mathrm{Bern}(x|\mu) = \mu^x(1-\mu)^{1-x}$$

  which is known as the Bernoulli distribution
- It is easily verified (Ex. 2.1) that $\mathbb{E}[x] = \mu$ and $\mathrm{var}[x] = \mu(1-\mu)$
- Now suppose we have a data set $\mathcal{D} = \{x_1, \ldots, x_N\}$ of observed values of a binary variable x, drawn independently from $p(x|\mu)$

Binary Variables

- We can construct the likelihood function of the date set as

$$p(\mathcal{D}|\mu) = \prod_{n=1}^{N} p(x_n|\mu) = \prod_{n=1}^{N} \mu^{x_n}(1-\mu)^{1-x_n}$$

- We can estimate μ by maximizing the log likelihood function

$$\ln p(\mathcal{D}|\mu) = \sum_{n=1}^{N} \ln p(x_n|\mu) = \sum_{n=1}^{N} \left\{x_n \ln \mu + (1-x_n)\ln(1-\mu)\right\}$$

  which depends on the observations x_n only through the sum $\sum x_n$
- This sum provides an example of a sufficient statistic for the data

Binary Variables

- Let the derivative of $\ln p(\mathcal{D}|\mu)$ with respect to μ be zero, so

$$\mu_{\mathrm{ML}} = \frac{1}{N}\sum_{n=1}^{N} x_n$$

  which is also known as the sample mean
- If we denote the number of observations of $x = 1$ within this data set by m, then

$$\mu_{\mathrm{ML}} = \frac{m}{N}$$

Binary Variables

- Now suppose we flip a coin 3 times and happen to observe 3 heads. Then $N = m = 3$ and $\mu_{\mathrm{ML}} = 1$.
- In this case, the maximum likelihood result would predict that all future observations should give heads.
- This is unreasonable, and it is an extreme example of the over-fitting associated with maximum likelihood

Binary Variables

- We can also find the distribution for the m observations of $x = 1$, given that a data size of N. This is called the binomial distribution.
- Out of N data points, we have to add up all of the possible ways of obtaining m heads

$$\mathrm{Bin}(m|N, \mu) = \binom{N}{m} \mu^m (1-\mu)^{N-m}$$

where the binomial coefficient

$$\binom{N}{m} \equiv \frac{N!}{(N-m)!m!}$$

is the number of ways of choosing m objects out of N identical objects (Ex 2.3)

Binary Variables

Example: choosing 2 objects from a set of 4 objects

- Let's take a simple example: choosing 2 objects from a set of 4 objects (say A, B, C, and D).
- Using the formula for the binomial coefficient:

$$\binom{4}{2} = \frac{4!}{(4-2)!2!} = \frac{24}{4} = 6$$

- So, there are 6 ways to choose 2 objects from 4.
- These distinct pairs are: (A, B), (A, C), (A, D), (B, C), (B, D), and (C, D).

Binary Variables

Example: Flipping a Biased Coin

- Suppose we flip a biased coin 5 times, where the probability of getting heads is $\mu = 0.6$ and tails is $1 - \mu = 0.4$.
- Find: the probability of getting exactly $m = 3$ heads in $N = 5$ flips
- The number of ways to get 3 heads in 5 flips is 10.
- The probability of getting exactly 3 heads is 0.6^3, and the probability of getting exactly 2 tails is 0.4^2
- So, the total probability of getting 3 heads and 2 tails is:

$$\mathrm{Bin}(3|5, 0.6) = \binom{5}{3} \cdot 0.6^3 \cdot 0.4^2 = 0.3456$$

- So, the probability of getting exactly 3 heads in 5 flips is 0.3456

Binary Variables

- For independent events, the mean of the sum is the sum of the means; the variance of the sum is the sum of the variances (Ex 1.10)
- Let x_i be the result of the i-th trial ($x_i = 1$ for a head; $x_i = 0$ for a tail)

$$m = x_1 + \cdots + x_N$$

- Each x_i follows a Bernoulli distribution with mean $\mathbb{E}[x_i] = \mu$.
- Therefore, the expected value of m is the sum of the expected values of the individual trials:

$$\mathbb{E}[m] = \mathbb{E}[x_1] + \mathbb{E}[x_1] + \cdots + \mathbb{E}[x_N]$$

Binary Variables

- So, the mean of the binomial distribution is:

$$\mathbb{E}[m] \equiv \sum_{m=0}^{N} m \cdot \mathrm{Bin}(m|N, \mu) = N\mu$$

- Since the variance of the sum is the sum of the variances

$$\mathrm{var}[m] \equiv \sum_{m=0}^{N} (m - \mathbb{E}[m])^2 \cdot \mathrm{Bin}(m|N, \mu) = N\mu(1-\mu)$$

Binary Variables

The beta distribution

- In the binomial distribution, the parameter μ is given by the fraction of the observations in the data set having $x = 1$
- As shown, this can give severely over-fitted results in the case when the sample size is small.
- In order to develop a Bayesian treatment for this problem, we introduce a prior distribution $p(\mu)$ over the parameter

Binary Variables

The beta distribution

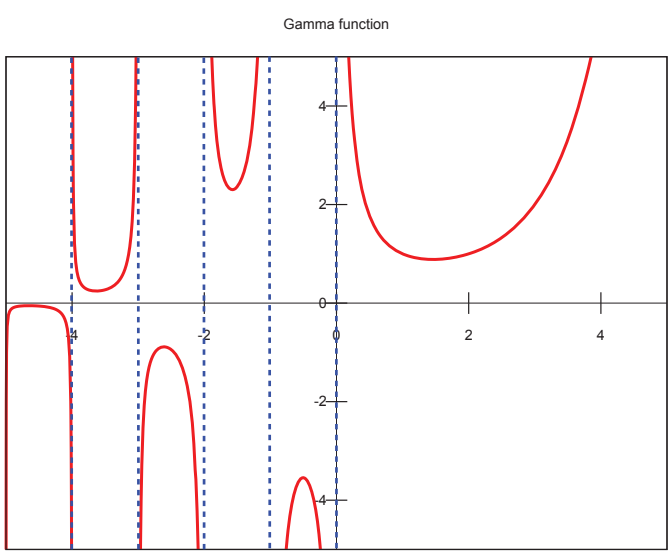
- To motivate this prior, we note that the likelihood function takes the form $\mu^x(1-\mu)^{1-x}$.
- We therefore choose a prior, called the beta distribution

$$\mathrm{Beta}(\mu|a,b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}\mu^{a-1}(1-\mu)^{b-1}$$

where $\Gamma(x)$ is the gamma function

$$\Gamma(x) = \int_0^\infty u^{x-1}e^{-u}\,\mathrm{d}u$$

Introduction to Probability Theory



- For Gamma function, we have

$$\Gamma(1) = 1$$
$$\Gamma(x+1) = x\Gamma(x)$$
$$\Gamma(x+1) = x!, \text{when } x \text{ is integer}$$

Binary Variables

- The coefficient in the beta distribution ensures that it is normalized,

$$\int_0^1 \mathrm{Beta}(\mu|a,b)\,\mathrm{d}\mu = 1$$

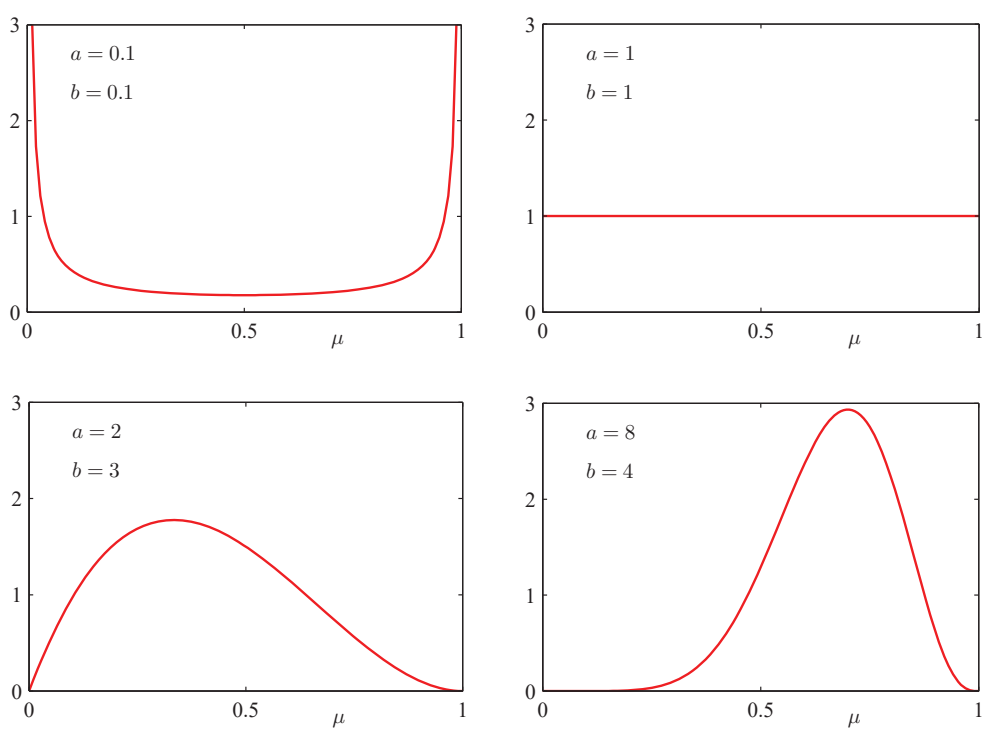
- The mean and variance of the beta distribution are given by

$$\mathbb{E}[\mu] = \frac{a}{a+b}$$
$$\mathrm{var}[\mu] = \frac{ab}{(a+b)^2(a+b+1)}$$

- The parameters a and b are often called hyperparameters because they control the distribution of the parameter μ

Binary Variables

Plots of the beta distribution with different values for a and b



Binary Variables

- The posterior distribution $p(\mu|\mathcal{D})$ of μ given data $\mathcal{D}$ is obtained by multiplying the beta prior by the binomial likelihood function

$$p(\mu|\mathcal{D}) \propto \mu^{a-1}(1-\mu)^{b-1}\mu^{m}(1-\mu)^{N-m} = \mu^{m+a-1}(1-\mu)^{l+b-1}$$

  where $l = N - m$

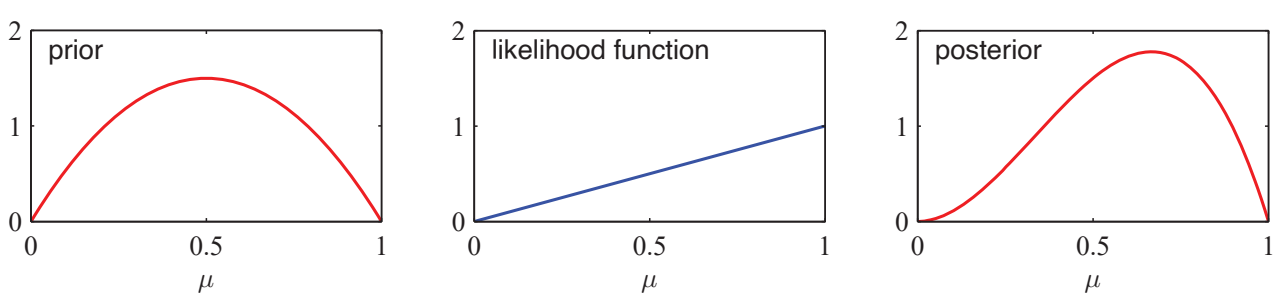
- Its normalization coefficient can therefore be obtained by comparison with beta distribution

$$p(\mu|\mathcal{D}) = \frac{\Gamma(m+a+l+b)}{\Gamma(m+a)\Gamma(l+b)}\mu^{m+a-1}(1-\mu)^{l+b-1}$$

- From the prior to the posterior distribution, we increased the a by m, and b by l

Binary Variables

- The posterior distribution can act as the prior if we subsequently observe additional data.
- We see that this sequential approach to learning arises naturally when we adopt a Bayesian viewpoint



Binary Variables

- To predict the outcome of the next trial, we must evaluate the predictive distribution of x, given the observed data $\mathcal{D}$
- From the sum and product rules of probability,

$$p(x=1|\mathcal{D}) = \int_0^1 p(x=1|\mu)p(\mu|\mathcal{D})\,\mathrm{d}\mu = \int_0^1 \mu p(\mu|\mathcal{D})\,\mathrm{d}\mu = \mathbb{E}[\mu|\mathcal{D}]$$

- Substitute the posterior distribution $p(\mu|\mathcal{D})$ into $p(x=1|\mathcal{D})$ yields

$$p(x=1|\mathcal{D}) = \frac{\Gamma(m+a+l+b)}{\Gamma(m+a)\Gamma(l+b)} \int_0^1 \mu^{m+a}(1-\mu)^{l+b-1}\,\mathrm{d}\mu$$

- The integral is known as the Beta function $B(m+a+1, l+b)$

Binary Variables

- Using the relationship between Beta and Gamma functions,

$$B(a,b) = \frac{\Gamma(a)\Gamma(+b)}{\Gamma(a+b)} \implies B(m+a+1, l+b) = \frac{\Gamma(m+a+1)\Gamma(l+b)}{\Gamma(m+a+1+l+b)}$$

- Substitute to $p(x=1|\mathcal{D})$ yields

$$p(x=1|\mathcal{D}) = \frac{\Gamma(m+a+l+b)}{\Gamma(m+a)\Gamma(l+b)} \frac{\Gamma(m+a+1)\Gamma(l+b)}{\Gamma(m+a+1+l+b)}$$

- Since $\Gamma(x+1) = x\Gamma(x)$, it simplifies to:

$$p(x=1|\mathcal{D}) = \frac{m+a}{m+a+l+b} = \frac{m+a}{N+a+b}$$

- In the limit of an infinitely large data set, it reduces to m/N

Outline

- Introduction
- Binary Variables
- Multinomial Variables
- Gaussian Distribution

Multinomial Variables

- Consider a K-dimensional vector $\mathbf{x}$ in which one of the elements equals 1, and all remaining elements equal 0.
- If $K = 6$ and $x_3 = 1$, then $\mathbf{x}$ will be represented by

$$\mathbf{x} = (0, 0, 1, 0, 0, 0)^{\mathrm{T}}$$

  Note that $\sum x_k = 1$
- If we denote the probability of $x_k = 1$ by the parameter μ_k, then the distribution of $\mathbf{x}$ is given

$$p(\mathbf{x}|\boldsymbol{\mu}) = \prod_{k=1}^{K} \mu_k^{x_k}$$

  where $\boldsymbol{\mu} = (\mu_1, \mu_2, \ldots, \mu_K)^{\mathrm{T}}$, and we have $\mu_k \geq 0$

Multinomial Variables

- Here $p(\mathbf{x}|\boldsymbol{\mu})$ is a generalization of the Bernoulli distribution to more than two outcomes
- This distribution is normalized over all possible $\mathbf{x}$

$$\sum_{\mathbf{x}} p(\mathbf{x}|\boldsymbol{\mu}) = \sum_{k=1}^{K} p(\mathbf{x} = \mathbf{e}_k|\mu) = \sum_{k=1}^{K} \mu_k = 1$$

where $\mathbf{e}_k$ is the one-hot vector with 1 in the k-th position and 0s elsewhere, and that

$$\mathbb{E}[\mathbf{x}|\boldsymbol{\mu}] = \sum_{\mathbf{x}} p(\mathbf{x}|\boldsymbol{\mu})\mathbf{x} = (\mu_1, \mu_2, \ldots, \mu_K)^{\mathrm{T}} = \boldsymbol{\mu}$$

Outline

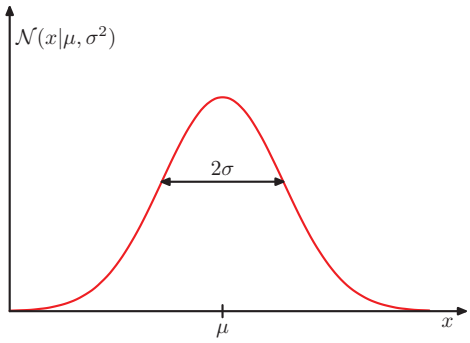
- Introduction
- Binary Variables
- Multinomial Variables
- Gaussian Distribution

Gaussian Distribution

- The Gaussian or normal distribution, is a widely used model for the distribution of continuous variables
- For a single variable x, the Gaussian distribution is given by

$$\mathcal{N}(x|\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}(x-\mu)^2\right\}$$

where μ is the mean and σ^2 is the variance.



Gaussian Distribution

- For a D-dimensional vector $\mathbf{x}$, the multivariate Gaussian distribution is

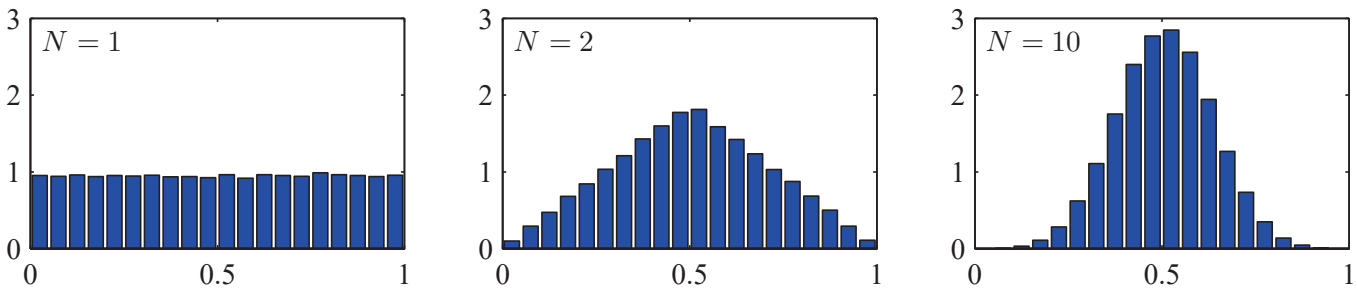
$$\mathcal{N}(\mathbf{x}|\boldsymbol{\mu},\boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{D/2}} \frac{1}{|\boldsymbol{\Sigma}|^{1/2}} \exp\left\{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\mathrm{T}}\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right\}$$

where $\boldsymbol{\mu}$ is a D-dimensional mean vector, $\boldsymbol{\Sigma}$ is a $D \times D$ covariance matrix, and $|\boldsymbol{\Sigma}|$ is the determinant of $\boldsymbol{\Sigma}$

- The Gaussian distribution arises in many different contexts and can be motivated from a variety of different perspectives

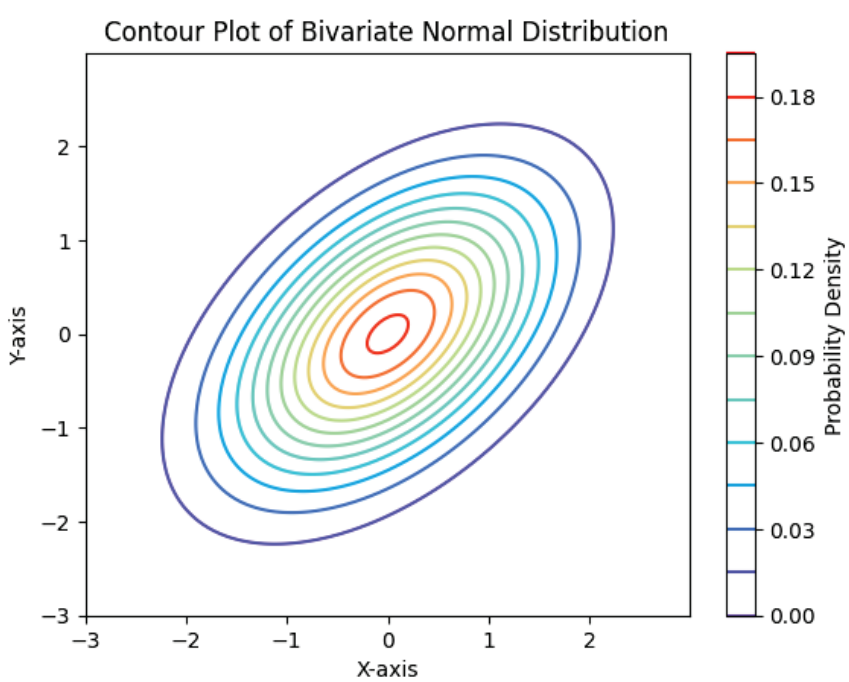
Gaussian Distribution

- Example: consider N variables $x_1, \ldots, x_N$ each of which has a uniform distribution over the interval $[0, 1]$
- Consider the distribution of the mean $(x_1 + \cdots + x_N)/N$. For large N, this distribution tends to a Gaussian



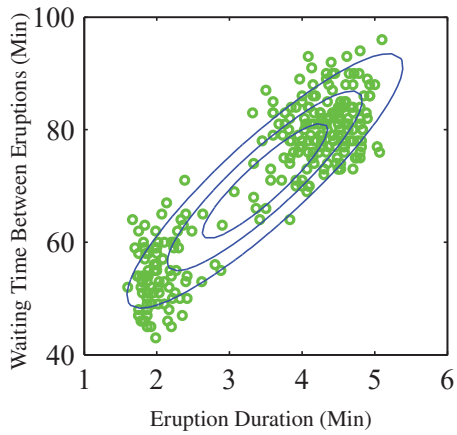
Marginal Gaussian distributions

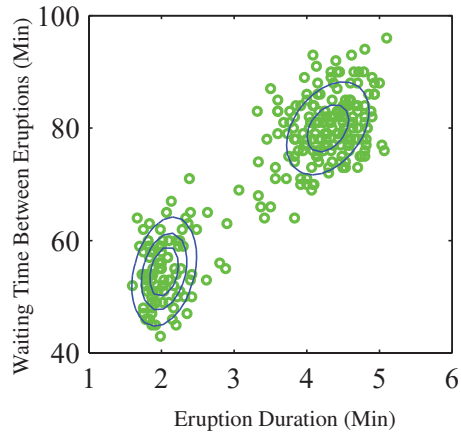
Contour plot of a multivariate Gaussian distribution of two variables (X,Y)



Mixtures of Gaussians

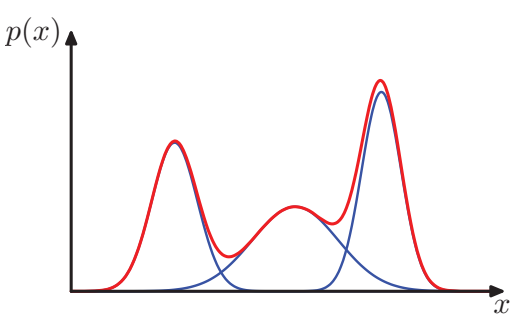
Gaussian distribution has important properties but also has significant limitations when modeling real data (Eruption of Old Faithful Geyser)





A linear superposition of two Gaussians gives a better characterization of the data than one Gaussian distribution

Mixtures of Gaussians



- Such superpositions can be formulated as probabilistic models known as mixture distributions
- Consider a superposition of K Gaussian distributions with mixing coefficients π_k

$$p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x|\mu_k, \sigma_k^2)$$

- By using a sufficient number of Gaussians, almost any continuous density can be approximated to arbitrary accuracy

Mixtures of Gaussians

- Each Gaussian distribution is called a component of the mixture distributions
- Now, integrating both sides of $p(x)$ with respect to x. Because both $p(x)$ and the individual Gaussians are normalized, we obtain

$$\int_{-\infty}^{\infty} p(x)\,\mathrm{d}x = \sum_{k=1}^{K}\left[\pi_k \int_{-\infty}^{\infty} \mathcal{N}(x|\mu_k, \sigma_k^2)\,\mathrm{d}x\right] = 1 \implies \sum_{k=1}^{K} \pi_k = 1$$

- Since $p(x) \geq 0$ and $\mathcal{N}(x|\mu_k, \sigma_k^2) \geq 0$, it implies $0 \leq \pi_k \leq 1$.